

Supply Chain Momentum Framework: Trading Manufacturing Dependencies

-Anthony Toscano

I propose a systematic trading framework that exploits a persistent market inefficiency: the idea that supplier stocks have a tendency to “lag” behind their customers’ market performances due to information delay in business-to-business (B2B) relationships. When companies report strong results that can be attributed to a specific product or material, the market will, in the short term, reprice these companies. However, the suppliers who are manufacturing these products or materials that lead to a company’s market success experience a delayed price appraisal. With this knowledge, we can use and exploit the relationship between buyer companies and supplier companies in order to make a signal whether to buy the supplier company based on the success of the buyer company in the market. For example, if Tesla, GM, and Ford are all seeing success, in the majority because of the move to electric vehicles (rather than this being a news or human-attributed market changer, I’m simply stating it as a fact of the natural evolution of technology), then we can expect a delayed jump in the suppliers of these EV batteries to these companies such as Panasonic/LG Energy.

Now, most investors typically analyze companies as isolated entities, examining their individual trends, thus the markets emotionally drive volatility in response to such factors. However, what is overlooked is the business-to-business relationship, which, if exploited correctly, can create predictable information about a supplier company that the market typically underprices. This exploitation, which makes opportunity both persistent and safe, comes in the form of a formula. This formula multiplies the amount of product purchased by a company of a supplier company's stock in relation to the other biggest companies that purchase the product. Then, multiplying these scaling percentages by the past 30-day performance of each of these top buyers, we are left with a number “x” which identifies the likelihood of a buy or a sell/hold. Then, adding a tolerance to this “x” of about $\pm 1\%$, we can come up with an effective strategy on whether to buy or short a supplier company based on the productivity of the buyer companies.

This strategy operates through three phases: relationship identification, signal construction, and position management. It begins with screening for suppliers exhibiting high customer concentration (companies that buy ~50% or more stock) with at least five years of clean price history to validate the relationship's predictive nature across multiple market cycles. Next comes signal construction. After the revenue-based weights for each major customer are calculated, we can use both a short-term (20-day) and a long-term (60-day) market cycle to help create a more refined signal (with 70% and 30% weighted, respectively, to favor responsiveness but hold sustainability). Finally comes the multi-layered filtering, which also allows for risk management. Using the current 20-day over the average of 60-day can calculate a volatility ratio for the supplier. If this ratio is over 1.5x, then reduce the position to 50%, which effectively maintains a partial exposure while protecting against false signals during times of instability. Additionally, a trend filter is also created, which calculates the supplier's 50-day moving average and longs only if our signal is above our 1% threshold and the price is above the moving average (short if vice-versa). With all of this, then comes the position management, specifically if all hold true or false (signal $\pm 1\%$ and both filters confirm or signal not $\pm 1\%$ and one or both filters reject), we allocate 100% of the capital to the trade or hold/reduce/exit in extreme imbalance.

These strategies try to minimize three main risks. The first one is correlation breakdown, specifically, if the customer company's success stops predicting the supplier's performance. This can be due to a multitude of factors, such as competitive changes or customer diversity, all culminating in the fact that our weights no longer have power as the purchasing of supplies is wrong. The way I intend to get around this issue is, as previously mentioned, the volatility filter, which will help better detect the formula's relationship becoming unstable and, in response, cut the position size when the correlation breaks. The second main risk is a sector-wide market crash in which the entire industry is failing due to macroeconomic forces being stronger than my relationship. In this case, the trend filter helps keep my strategy neutral during these downturns as we only trade long when the supplier price is above the 50-day moving average. Additionally, helps our maximum drawdown. The last risk this strategy can detect and fix is false signals, which have already been extensively discussed in previous paragraphs.

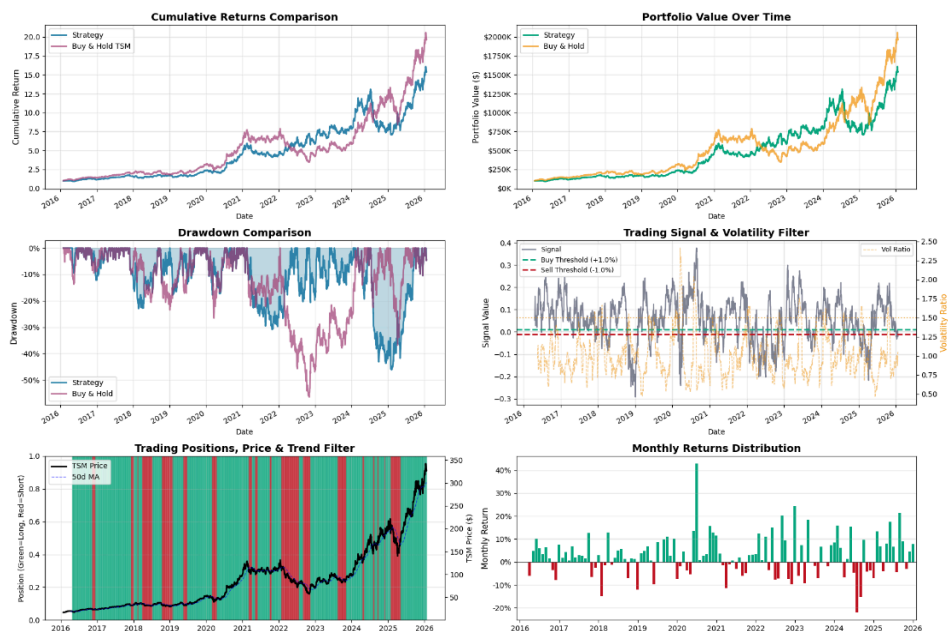
My strategy only trades large, highly liquid stocks where both the supplier company and its buyers have \$100 million or more in daily volume, so as not to move the market with my trades. Due to my very selective filters working in correspondence with the formula, my strategy is on the more selective side, opting for fewer trades per year (in turn keeping transaction costs low). The individual supplier-customer relationship can handle \$50-100 million in capital before my strategy becomes rather unpredictable. I believe the greatest scalability of my strategy comes with the numerous supplier-buyer relationships that exist in the market that can be exploited [EV's, semiconductors, pharmaceuticals, etc.]. This, paired with the low trading frequency, allows this strategy to grow much larger before hitting eventual "unpredictable" limits.

To demonstrate my strategy in actual I made a 10-year simulation of it running on Taiwan Semiconductor Manufacturing. TSM is the premier manufacturer of advanced AI chips and semiconductors. The top 3 companies that buy supply from TSM are Apple (22.5%), NVIDIA (11.5%), and AMD (9.5%). With these percentages in mind, we can use them as scalars in relation to the respective company's performance. When we run this with our volatility and trend filters over a 10 year period (~2016~2026) with an initial capacity of \$100,000 we see a total strategy return of 1438.53% (annualized ~31.54%). What's interesting about my strategy when compared to just buying and holding the stock is that it almost perfectly mimics returns (slightly below, but as to be expected from a low-risk strategy and this specific example), but beats its max drawdown by 10% (my strategy:-46.16% buy/hold:-56.47%). Not to mention my volatility is about 2% smaller than the standard strategy, with a profit factor of 1.2 and 72 trades. Granted, this is a very beginner strategy, kinned to my experience in this type of development (my win rate sits a little over 50%), but for it to almost directly mimic the regular buy and hold strategy while outperforming in general risk management says a lot about 100% expected long-term profit for my strategy. Again, scaling this type of strategy to other B2B relationships could yield an extremely low-risk long-term profit portfolio.

Code: [Click here](#)

DATA FOR TSM B2B EXAMPLE:

BACKTEST PERFORMANCE METRICS	
Total Strategy Return.....	1438.53%
Total Buy & Hold Return.....	1868.66%
Annualized Strategy Return.....	31.54%
Annualized B&H Return.....	34.83%
Strategy Sharpe Ratio.....	1.02
B&H Sharpe Ratio.....	1.06
Strategy Max Drawdown.....	-46.15%
B&H Max Drawdown.....	-56.47%
Strategy Volatility.....	31.90%
B&H Volatility.....	33.53%
Win Rate.....	52.38%
Profit Factor.....	1.20
Number of Trades.....	72
Total Transaction Costs..... \$	0.09
Final Portfolio Values.....	
Strategy: \$15,385,341.71	
Buy & Hold: \$19,686,637.78	
Difference: \$-4,301,296.07 (-21.85%)	
COMPLETE	



$$x = (std : percentBuyAPPL) * (AAPL30DayPerf.) + (std : percentBuyNVDA) * (NVDA30DayPerf.) + (std : percentBuyAMD) * (AMD30DayPerf.)$$